ILMINGTON UNIVERSITY

COLLEGE OF TECHNOLOGY

SEC 210 – Principles and Practice of Information Security

11/22/2020	Tyler Sanderson

OBJECTIVES: In this lab you will run the **Kali Linux** OS on the **CNSLabSvrDC** VM. Kali Linux is a collection of ethical hacking tools. During the course, you will use some of them to demonstrate their capabilities. This is NOT found in the **Lab Manual**.

INSTRUCTIONS: Carefully follow the instructions on each page. You will submit several screenshots. Search for INSERT or SCREENSHOT in uppercase to find out the exact pages. There are questions to answer. Search for QUESTION in uppercase to find out the exact pages.

PREREQUISITES:

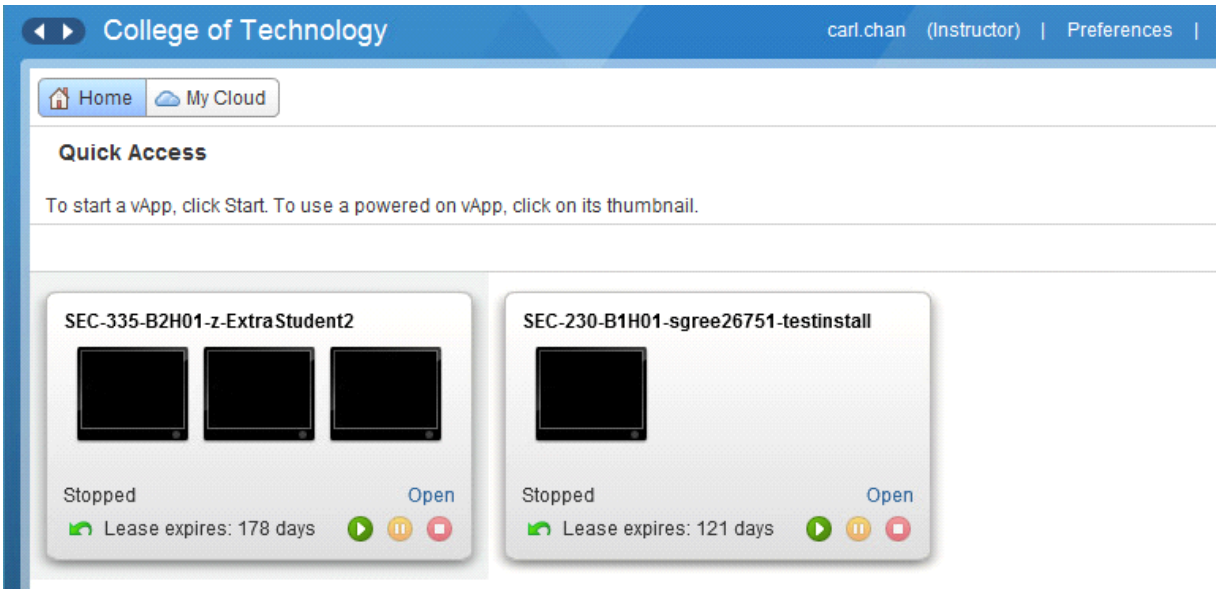
- "How to use the Snipping Tool on Windows 10" video at https://www.youtube.com/watch?v=ayi5-7QPNcA
- "How to Take a Screenshot on a Mac" video at https://www.youtube.com/watch?v=h3RkDPfphSQ
- How to access SEC 210 vApp.docx

COMMENTS: I do NOT want a screenshot of your entire VM desktop window when only a portion is needed to demonstrate that you have followed the instructions. For examples, see the screenshots in this document.

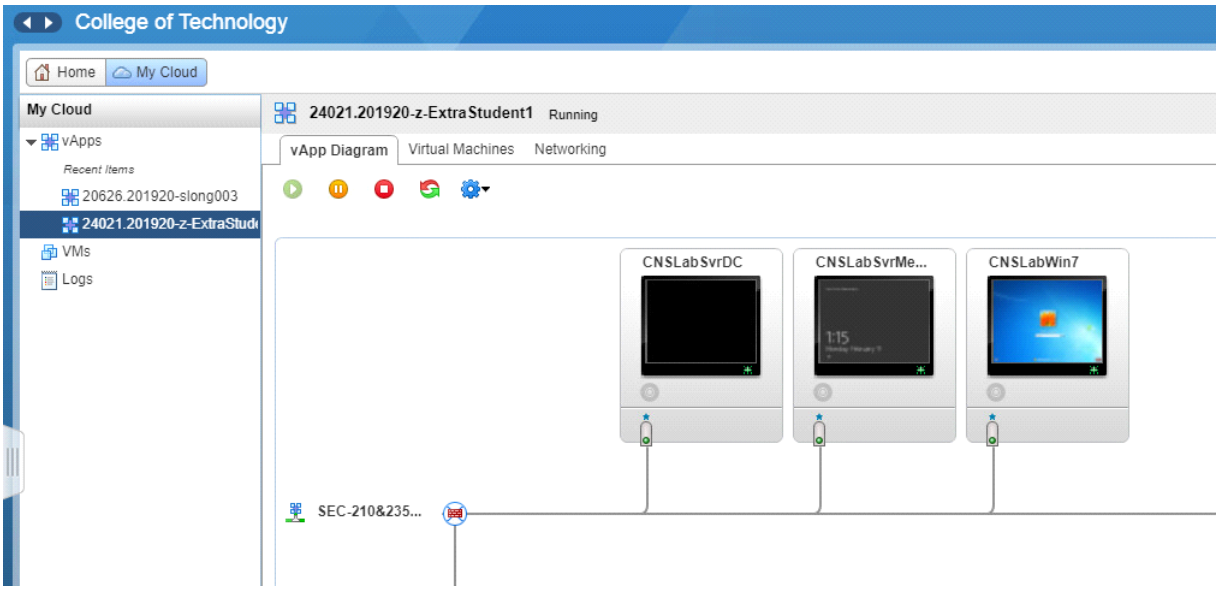
Ensure that you complete the entire lab. You will know when you see "THIS IS THE LAST PAGE OF THE LAB

Inserting a DVD

- After logging into vCloud, select your vApp. NOTE: You will not see the same title shown below. Your title should have the form of <CRN>.<Term>-<Student ID>; e.g. 24021.201902-johndoe001
- Place your mouse cursor on a blank area, not on one of the VM icons
- **Double-click** on the vApp

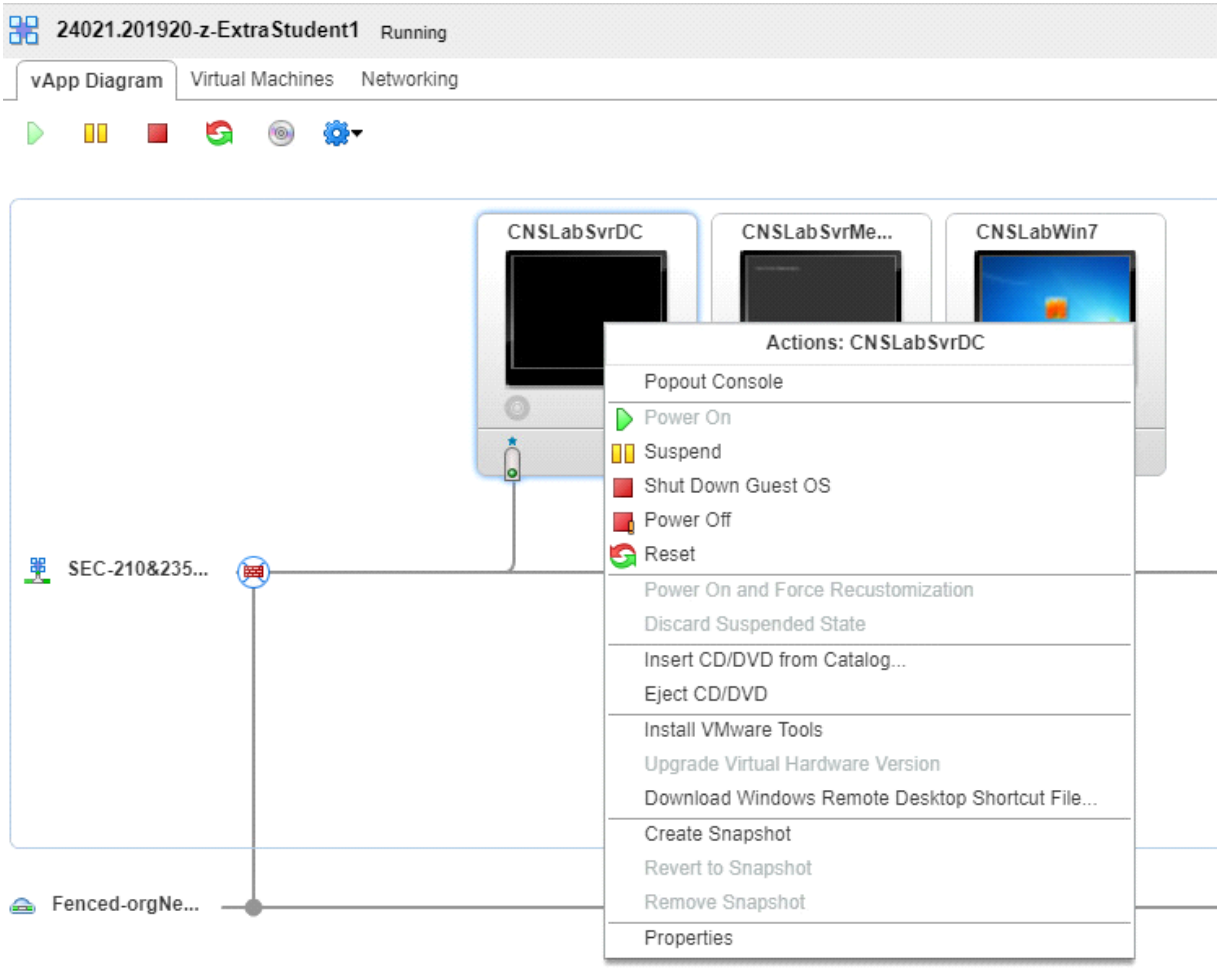


- vApp Diagram folder tab window appears. The **My Cloud folder** tab is now highlighted

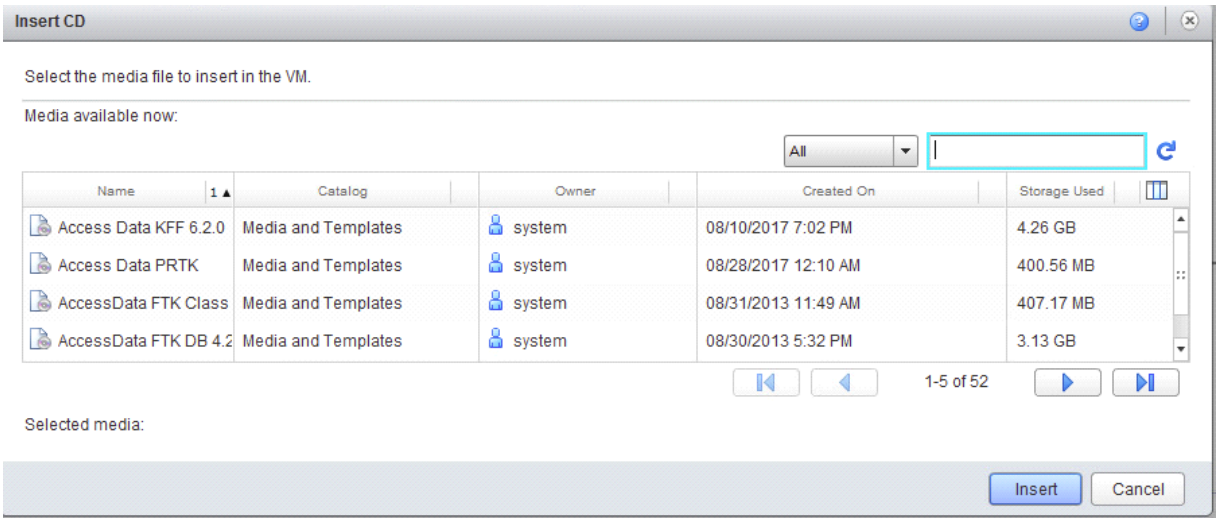


- Hover mouse cursor over **CNSLabSvrDC**, the left-most VM
- NOTE: The circle is greyed out indicating that no DVD has been inserted
- Right-click mouse, pop-up appears

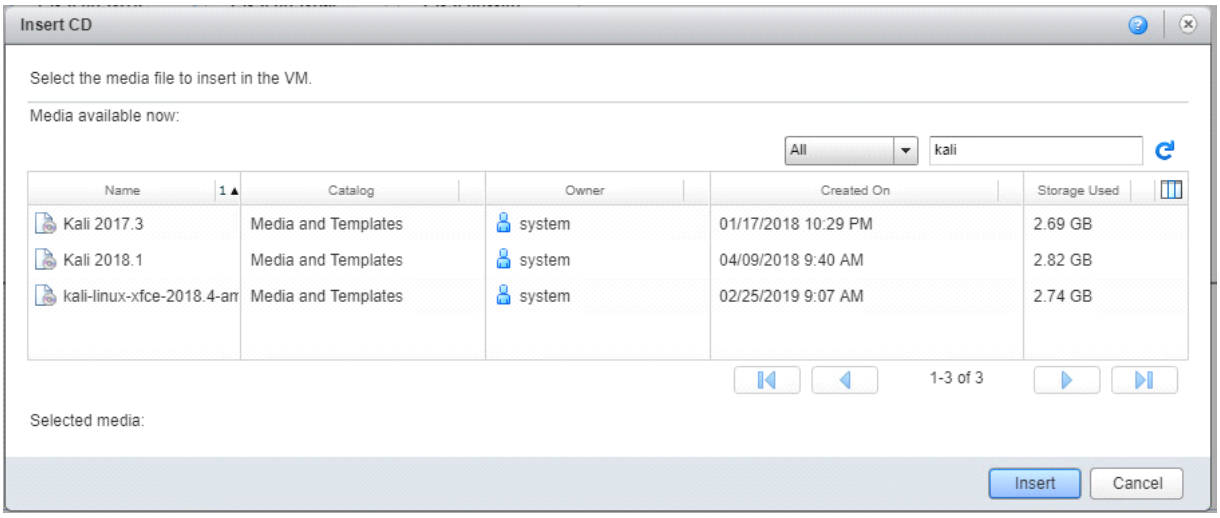
- Select **Insert CD/DVD from Catalog**



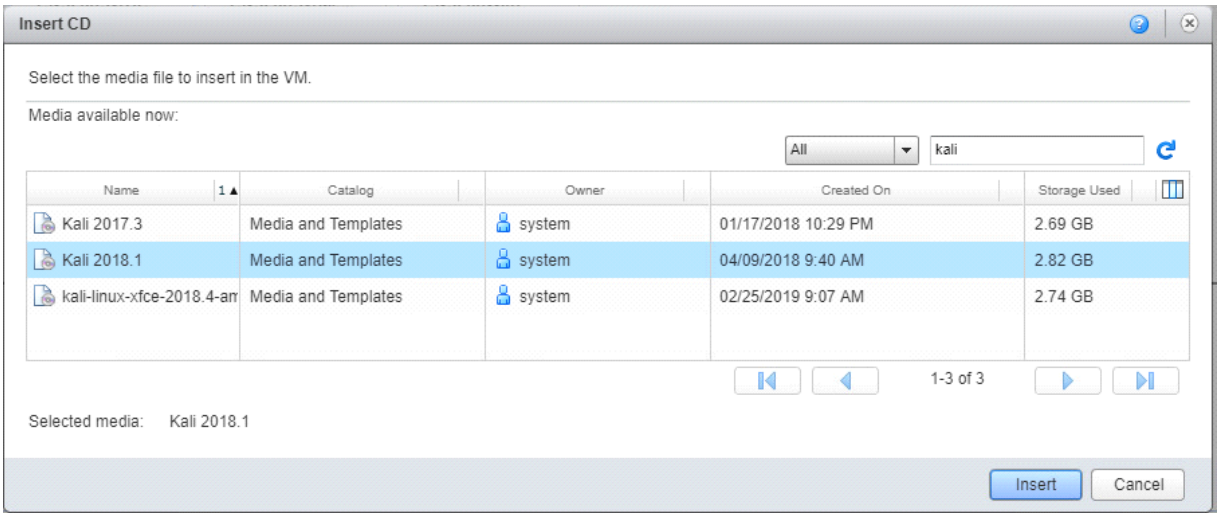
- The **Insert CD** window appears



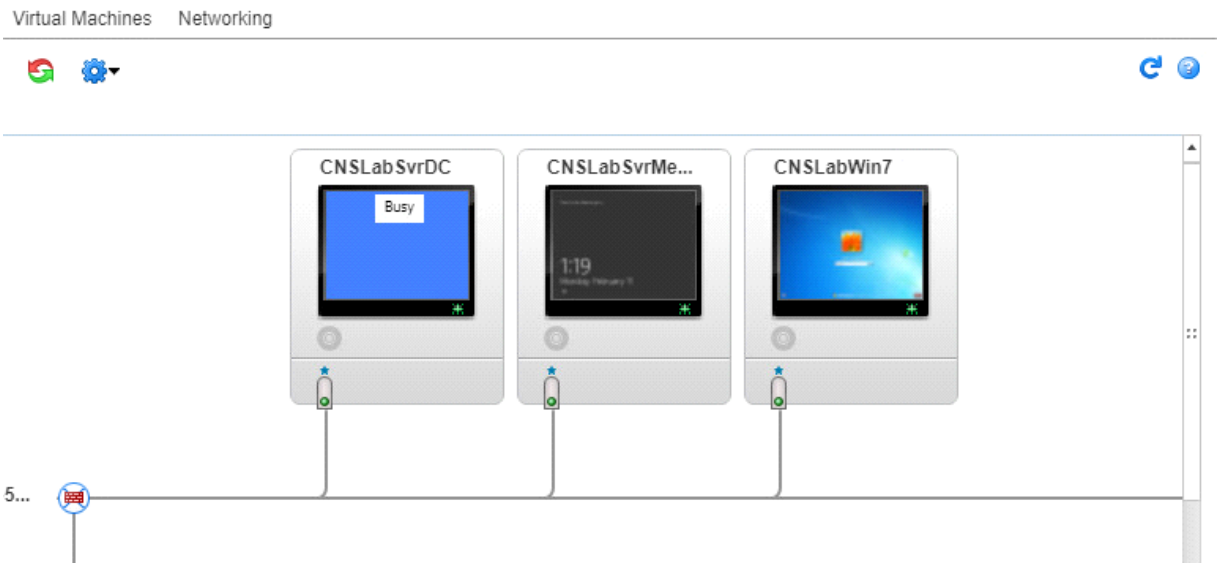
- Enter **kali** and press **Enter**
- The window now shows several **kali** entries



- Select the **Kali 2018.1** entry to highlight it in Blue
- Click on **Insert**

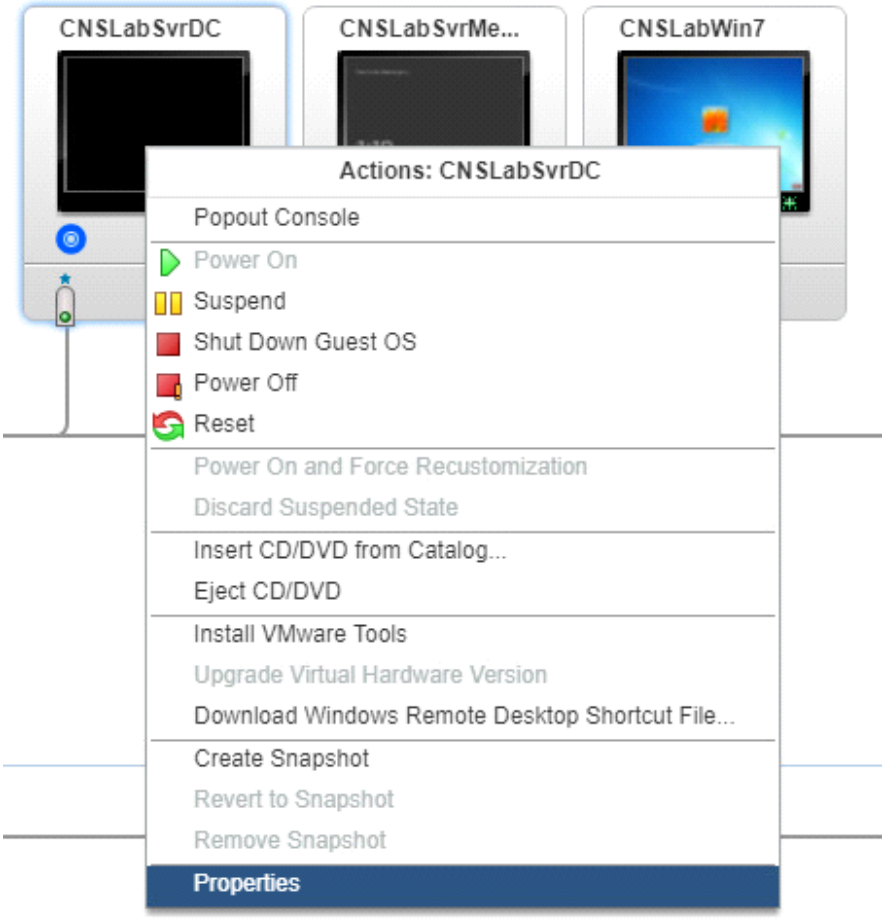


- **Busy** appears in the **CNSLabSvrDC** icon
- Click on **Refresh** icon

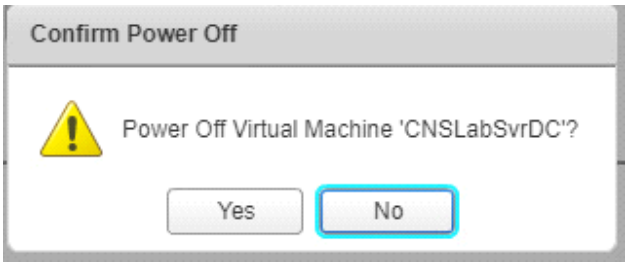


- **Busy** goes away
- NOTE: The circle is now blue indicating that a DVD has been inserted
- Hover mouse cursor over **CNSLabSvrDC**
- Right-click the mouse. A pop-up appears

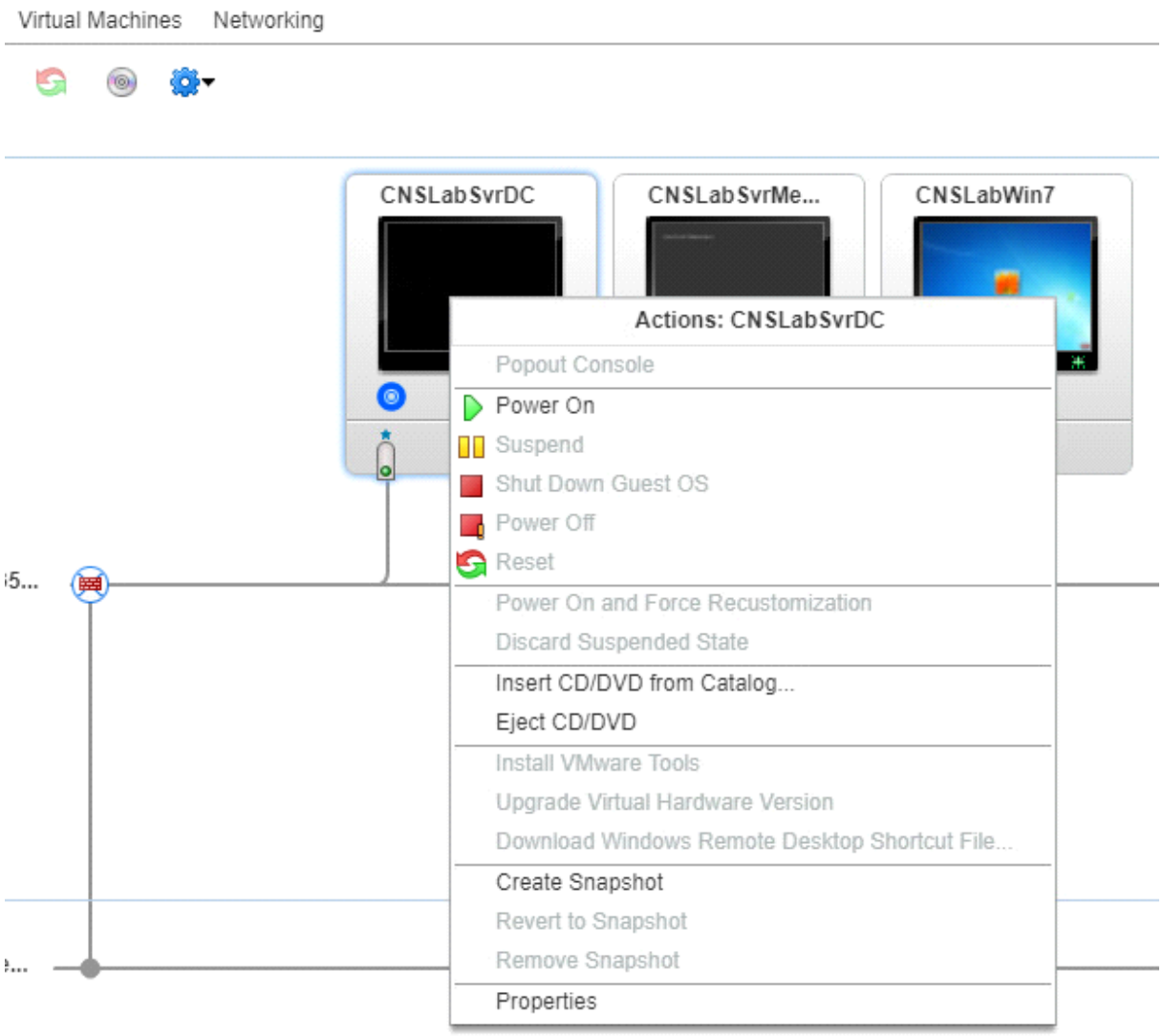
- Click on **Power Off**.



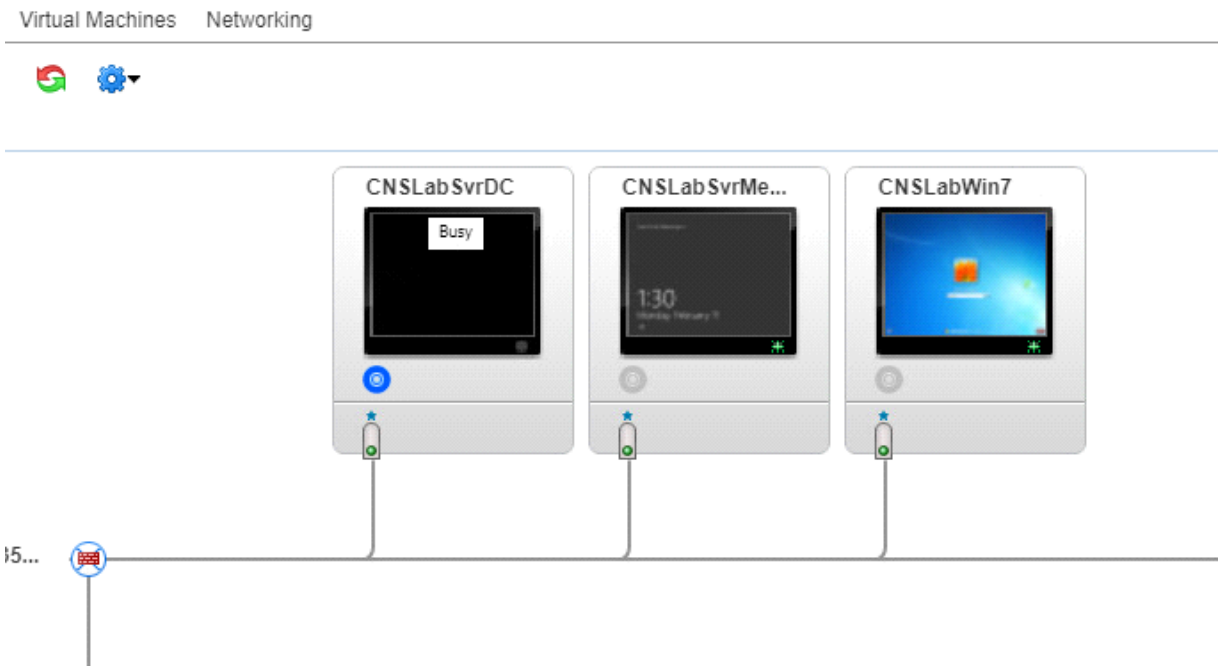
- You will see the following window. Click on **Yes**
- **Busy** appears but goes away after a short while



- Hover mouse cursor over **CNSLabSvrDC** again and right-click the mouse.
- Click on **Power On**



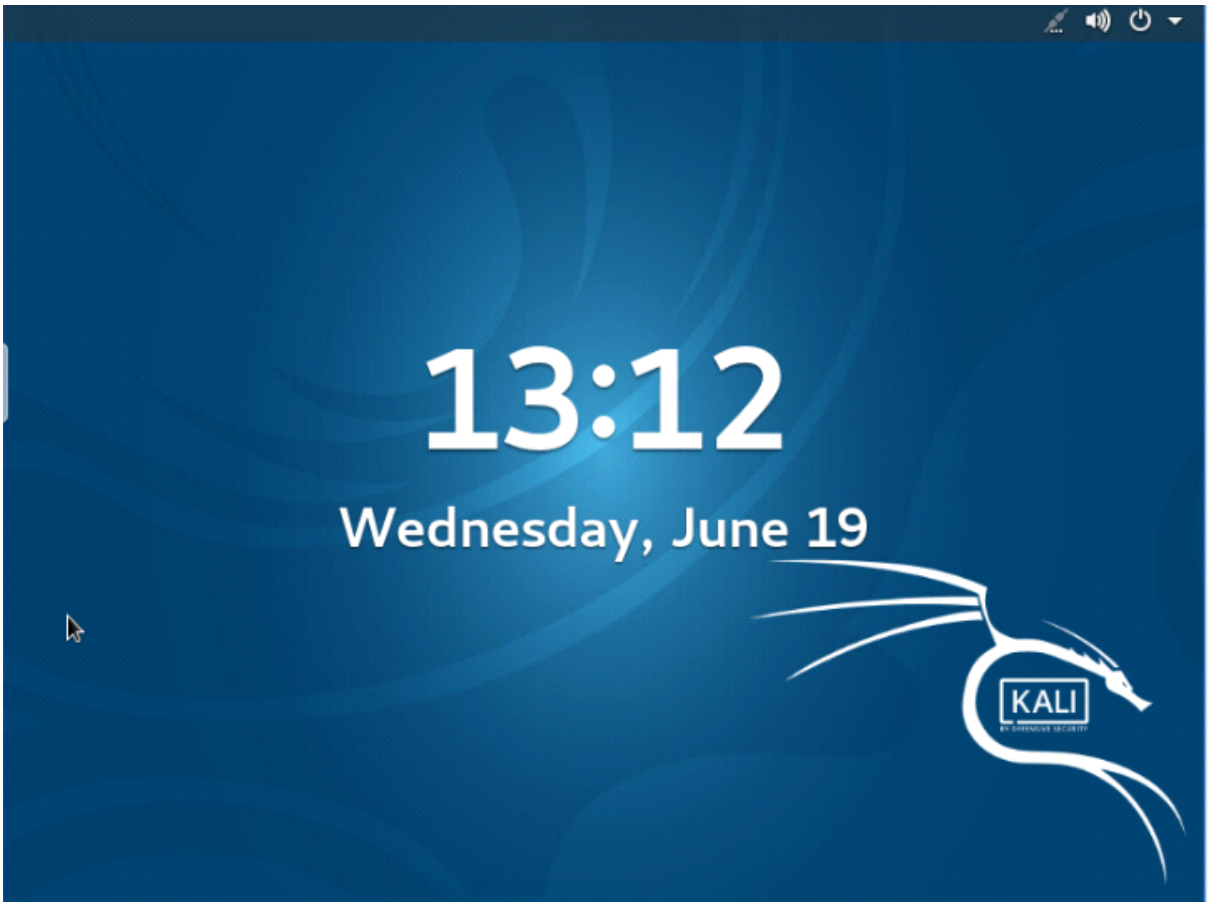
- **Busy** appears again
- After a while **Busy** goes away on its own
- Place the mouse cursor on the black area of the VM, not the blank area
- **Double-click** on **CNSLabSvrDC**



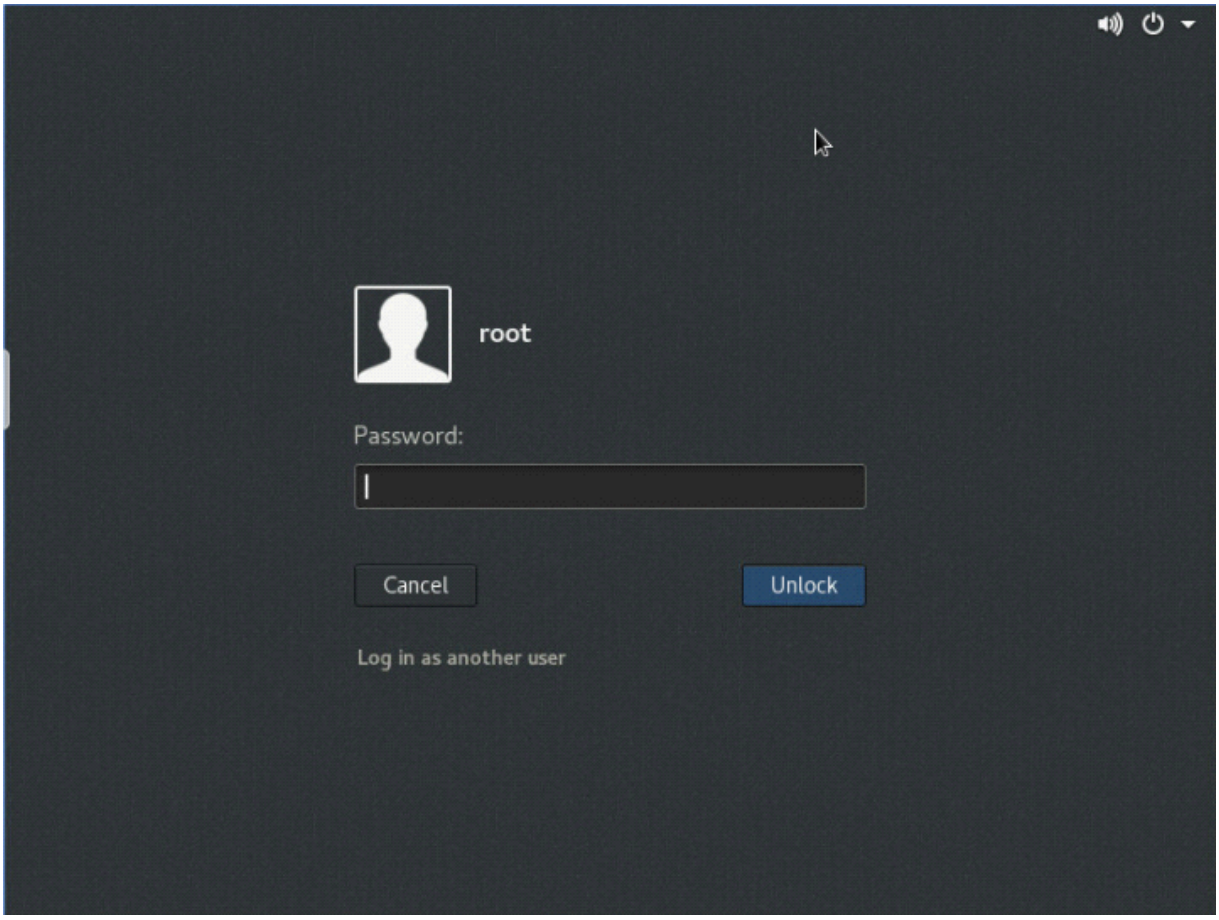
Starting up Kali

- The following startup screen appears
- Resize the window as needed
- Click the mouse cursor inside the window
- **Live (amd64)** should already by selected
- Press **Enter**

- Eventually, the following screen appears
- Place the mouse cursor in the window
- Left-click the mouse

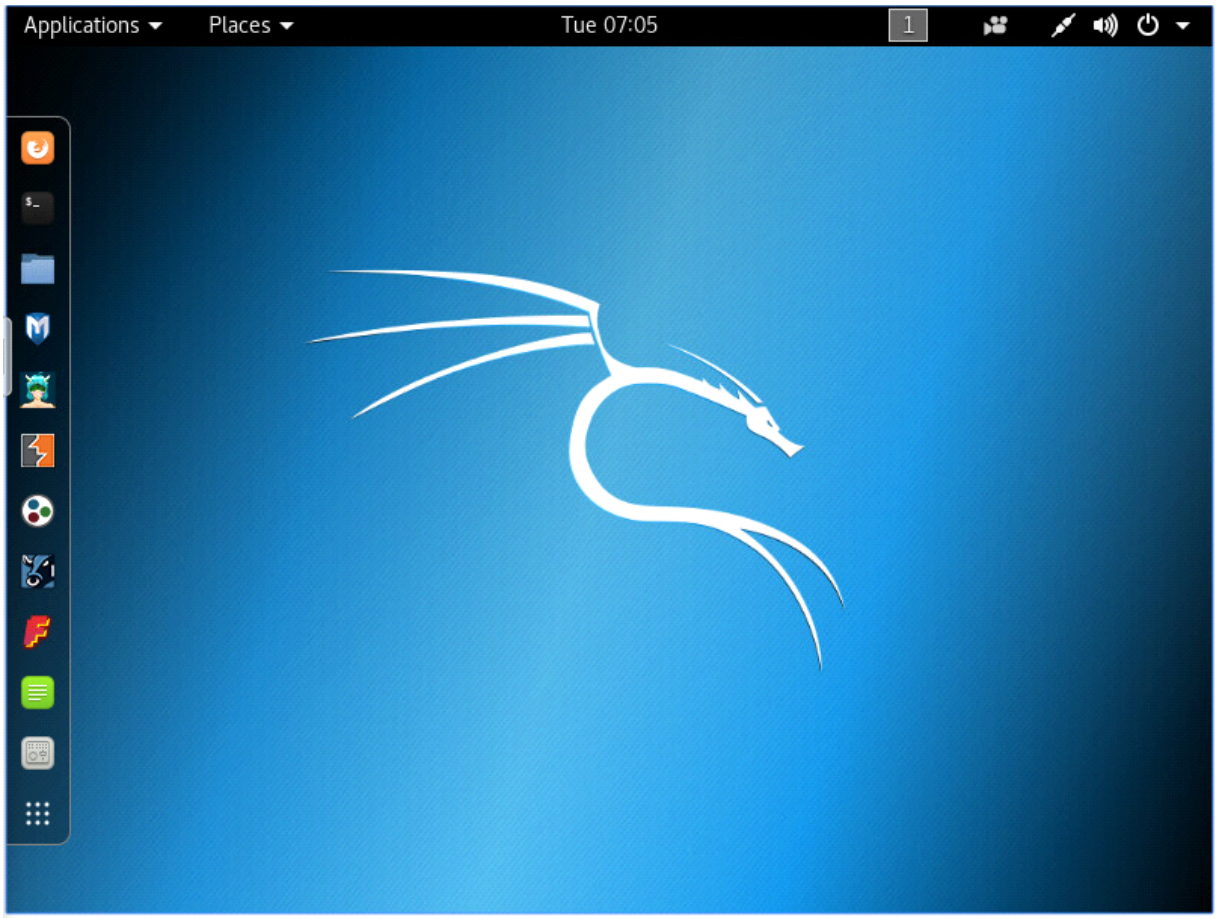


- Press **Enter**
- The login screen appears

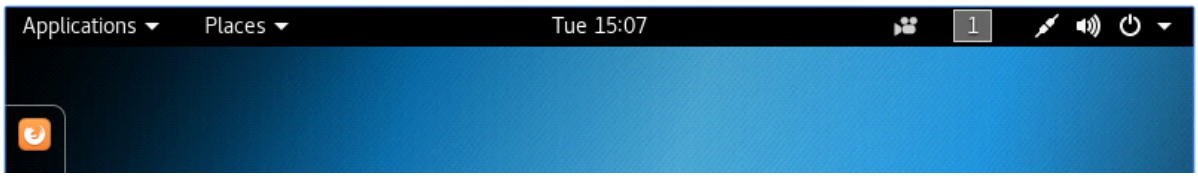


- Enter **toor** as the password; i.e. **root** spelled backwards
- Press **Enter**

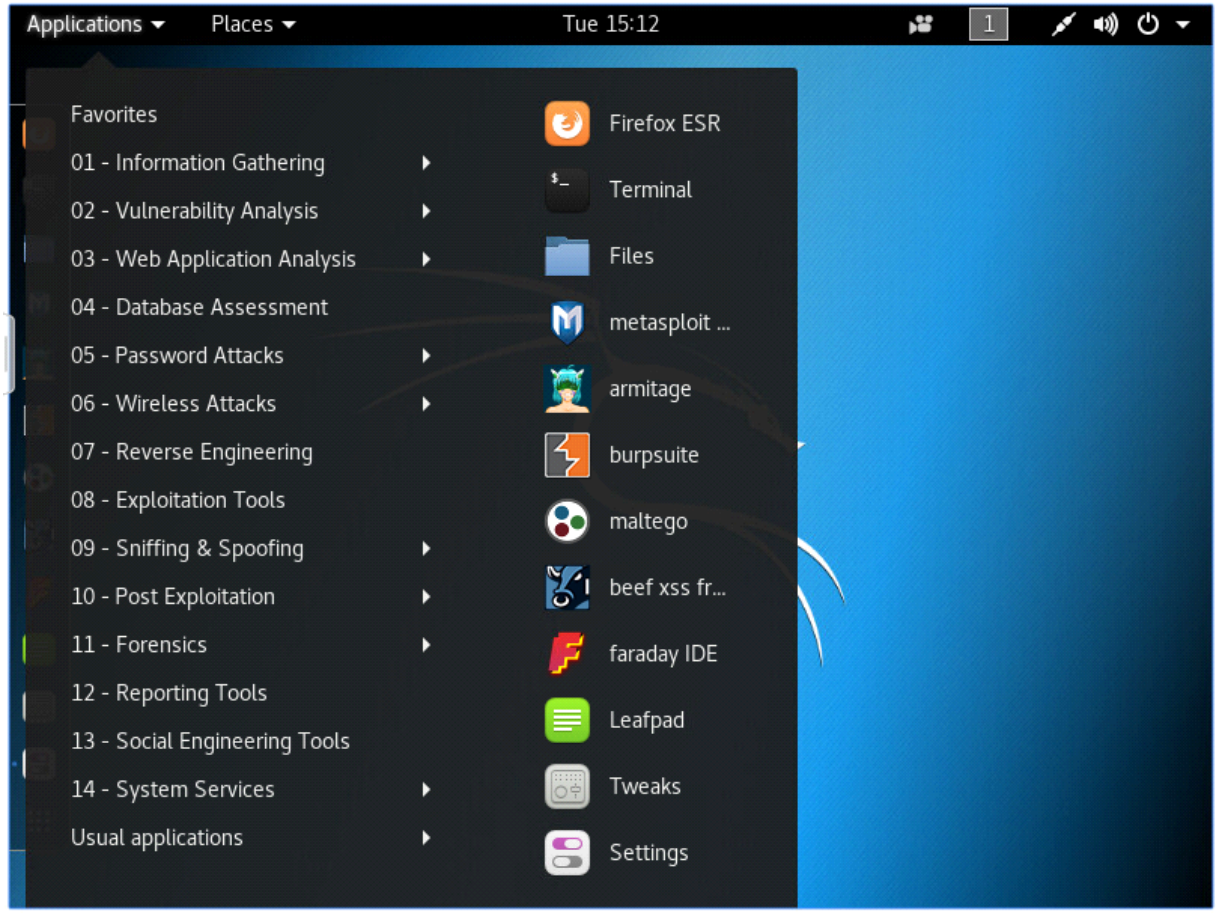
- Tthe Kali graphical desktop appears
- Click on the second icon on the left to open a terminal window



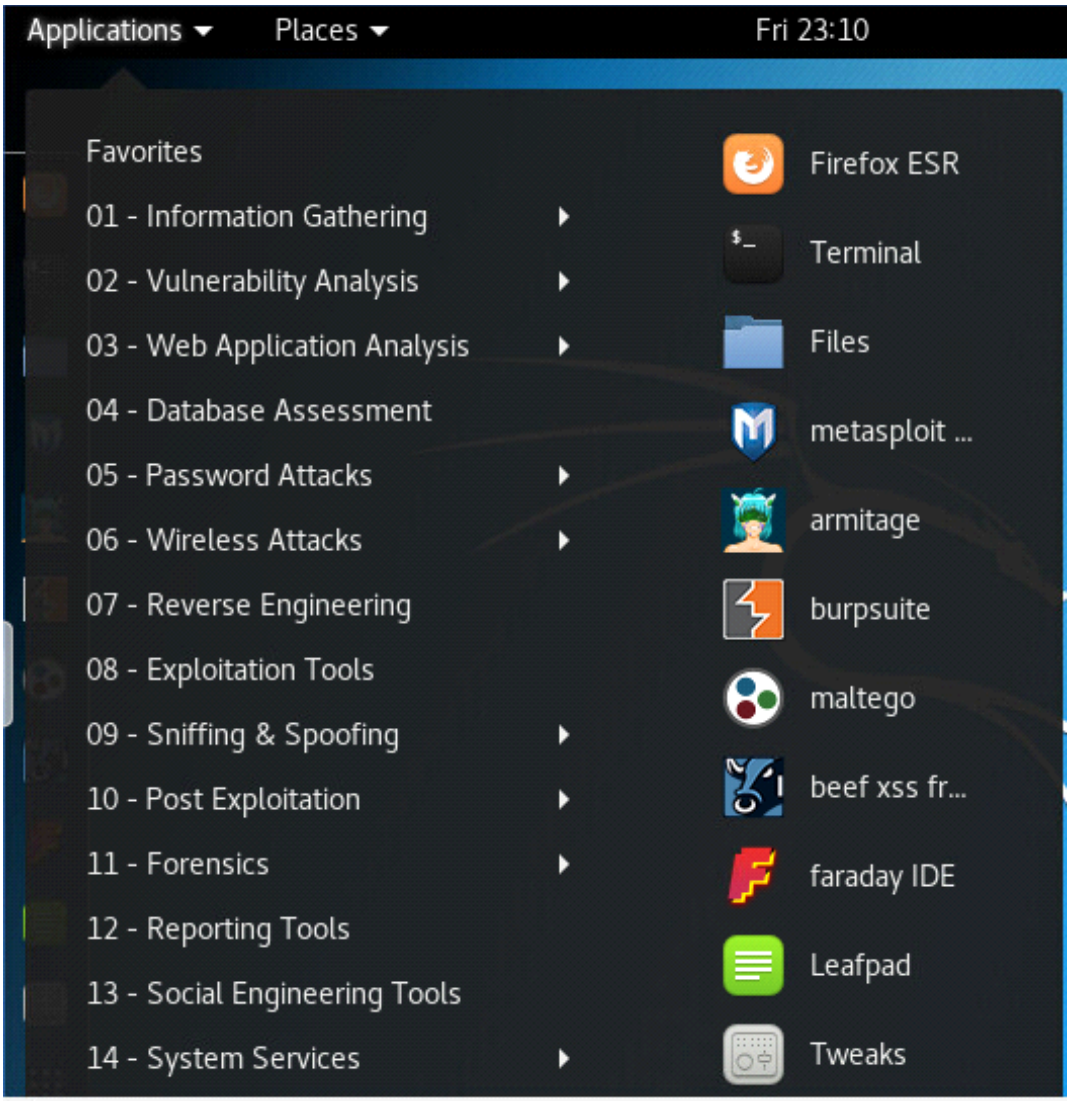
- Click on **Applications** in the top left corner



- A menu appears showing a category list of hacking tools
- Take a screenshot that includes the day and time in the top bar. It should look like



INSERT YOUR SCREENSHOT HERE. IT IS WORTH ***5 points***



You will be asked questions about the tools available in Kali. I want you to write out the answers. Do NOT just take a screenshot

Information Gathering Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Information Gathering**. List the tools shown on the right.

Some tools shown to the right inside 'Information Gathering' are dmitry, dnmap-cli..., dnmap-ser..., ike-scan, maltego, netdiscover, nmap, p0f, recon-ng, sparta, and zenmap.

This QUESTION is worth ***5 points***. Write your answer in the box

Click on **Information Gathering**. List the three sub-categories shown.

Information lists DNS Analysis, IDS/IPS Identification, and Live Host Identification as sub-categories.

This QUESTION is worth ***5 points***. Write your answer in the box

Highlight the **first** subcategory under **Information Gathering**. List the tools shown on the right

To the right of DNS Analysis, the first sub-category in the list, are dnsenum, dnsmap, dnsrecon, dnstracer, dnswalk, fierce, and urlcrazy.

Vulnerability Analysis Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Vulnerability Analysis**. List the tools shown on the right.

Next to Vulnerability Analysis are golismero, lynis, nikto, nmap, sparta, and unix-prives...

This QUESTION is worth ***5 points***. Write your answer in the box

Click on **Vulnerability Analysis**. List the three sub-categories shown.

The three subcategories of Vulnerability Analysis include Cisco Tools, Fuzzing Tools, and Stress Tools.

This QUESTION is worth ***5 points***. Write your answer in the box

Highlight the **second** subcategory under **Vulnerability Analysis**. List the tools shown on the right

The tools under the second subcategory, Fuzzing Tools, include bed, ohrwurm, powerfuzzer, sfuzz, siparmyknife, spike-gene..., spike-gene..., spike-gene..., and spike-gene...

Password Attacks Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Password Attacks.**. List the tools shown on the right

The tools listed to the right of Password Attacks are cewl, crunch, hashcat, john, johny, medusa, ncrack, ophcrack, pyrit, rainbowcra..., rcracki_mt, and wordlists.

This QUESTION is worth ***5 points***. Write your answer in the box

Click on **Password Attacks**. List the three sub-categories shown

The three subcategories shown for Password Attacks are Offline Attacks, Online Attacks, and Passing the Hash tools.

This QUESTION is worth ***5 points***. Write your answer in the box

Highlight the **third** subcategory under **Password Attacks**. List the tools shown on the right

The third subcategory, Passing the hash tools, lists mimikatz, pth-curl, pth-net, pth-openc..., pth-rpcclient, pth-smbcli..., pth-smbget, pth-sqsh, pth-winexe, pth-wmic, pth-wmis, and smbmap.

Exploitation Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Exploitation.**. List the tools shown on the right

The tools left of Exploitation tools are armitage, beef xss fr..., metasploit..., msf payloa..., searchsploit, social engi..., sqlmap, and termineter

Social Engineering Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Social Engineering.**. List the tools shown on the right

The tools listed right of Social Engineering Tools are backdoor-f..., beef xss fr..., ghost phis..., maltego, msf payloa..., social engi..., and u3-pwn.

Sniffing & Spoofing Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Sniffing & Spoofing.**. List the tools shown on the right

The tools listed to the right of Sniffing & Spoofing are bdfproxy, driftnet, ettercap-g..., hamster, macchanger, mitmproxy, netsniff-ng, responder, and wireshark.

This QUESTION is worth ***5 points***. Write your answer in the box

Click on **Sniffing & Spoofing**. List the two sub-categories shown.

The subcategories of Sniffing & Spoofing are Network Sniffers and 'Spoofing and MITM'.

This QUESTION is worth ***5 points***. Write your answer in the box

Highlight the **first** subcategory under **Sniffing & Spoofing**. List the tools shown on the right

The tools listed under the first subcategory of Sniffing & Spoofing, Network Snifferes, are darkstat, dnschef, dsniff, hexinject, netsniff-ng, nfspy, sslsniff, and tcpflow.

Post Exploitation Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Post Exploitation.**. List the tools shown on the right

The tools listed to the right of Post Exploitation are backdoor-f..., bdfproxy, exe2hex, intersect, mimikatz, nishang, powersploit, proxychains, and weevely.

This QUESTION is worth ***5 points***. Write your answer in the box

Click on **Post Exploitation**. List the three sub-categories shown

The three subcategories of Post Exploitation are OS Backdoors, Tunneling & Exfiltration, and Web Backdoors.

This QUESTION is worth ***5 points***. Write your answer in the box

Highlight the **second** subcategory under **Post Exploitation**. List the tools shown on the right

The tools listed to the right of the second subcategory of Post Exploitation, Tunneling & Exfiltration, are dbd, dns2tcpc, dns2tcpd, exe2hex, iodine, miredo, proxychains, proxytunnel, ptunnel, pwnat, sslh, and stunnel4. There is another tool but I cannot resize the window to see it.

Forensics Tools

This QUESTION is worth ***5 points***. Write your answer in the box

Place the mouse cursor on **Forensics.**. List the tools shown on the right

The tools listed to the right of Forensics are autopsy, binwalk, bulk_extra..., chkrootkit, foremost, galleta, hashdeep, volafox, and volatility.

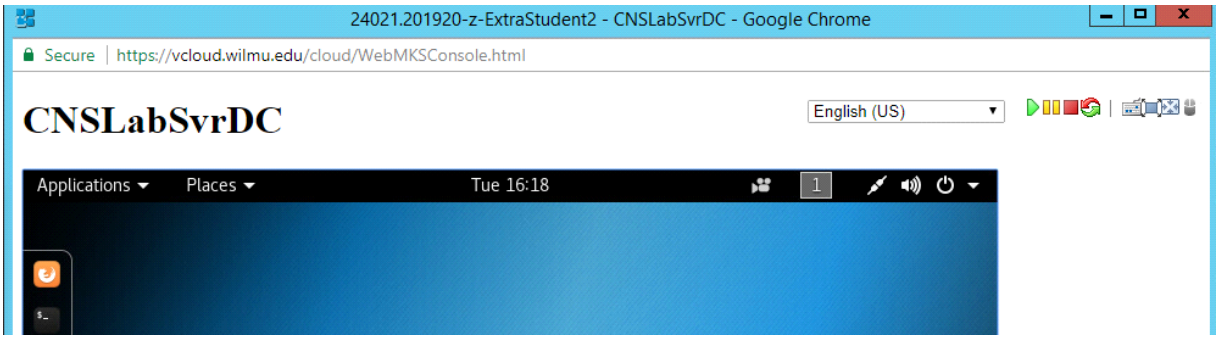
This QUESTION is worth ***5 points***. Write your answer in the box

Click on **Forensics**. List the three sub-categories shown

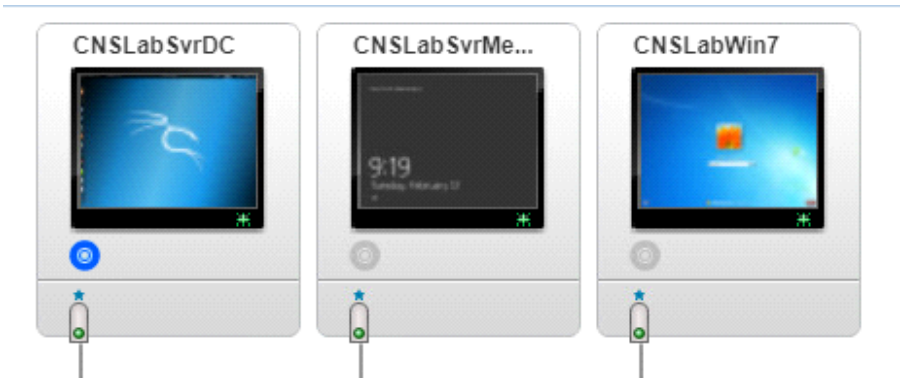
The three subcategories of Forensics are Digital Forensics, Forensic Carving Tools, and Forensic Imaging Tools.

Ejecting a DVD

- Exit from the **CNSLabSvrDC** window by clicking on the **X** in the top-right corner

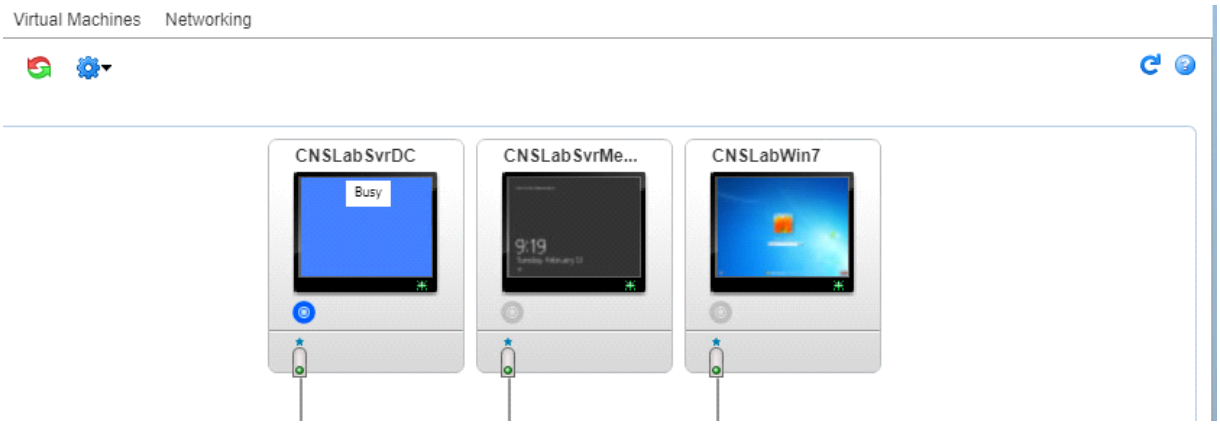


- You should return to the vApp diagram



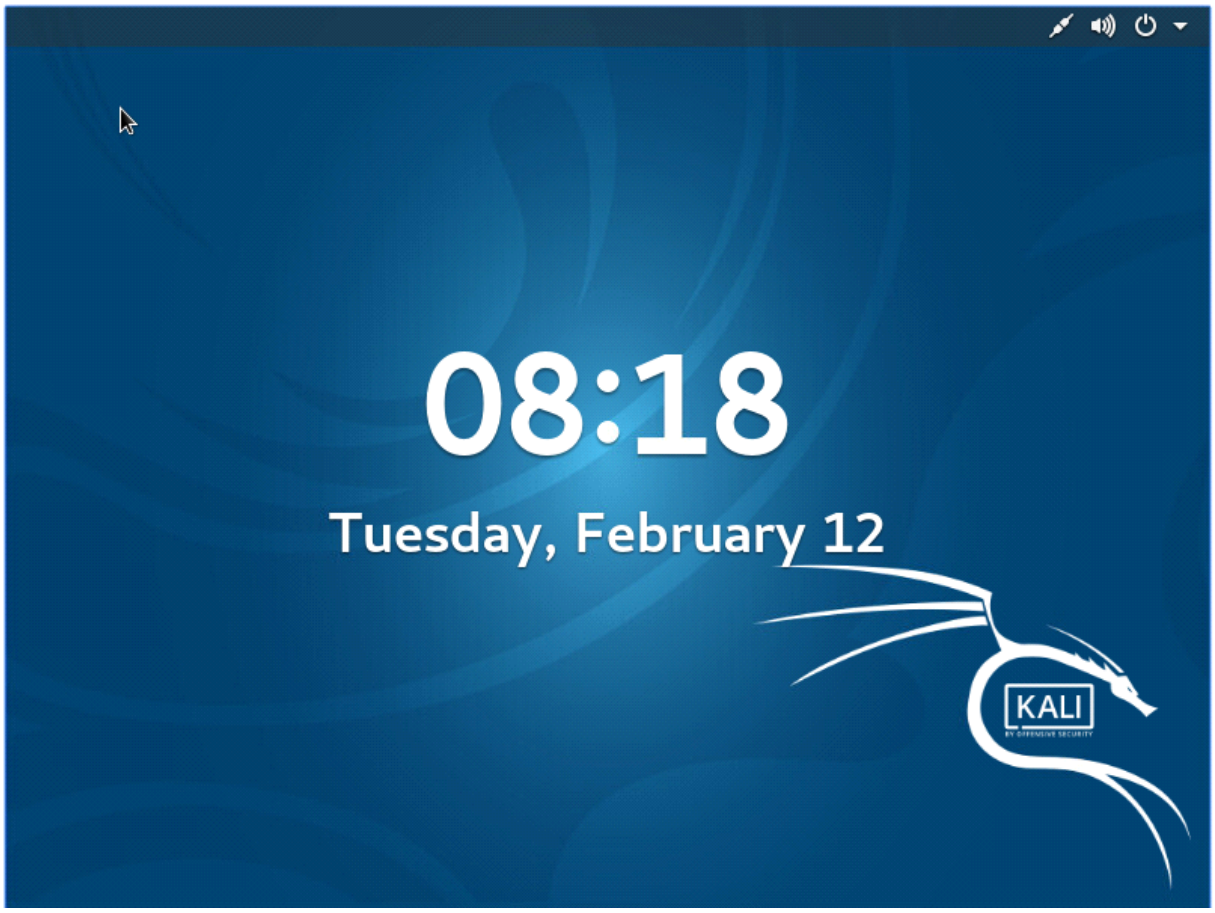
- Hover mouse cursor over **CNSLabSvrDC**
- Right-click the mouse. A pop-up appears
- Select **Eject CD/DVD** and press **Enter**
- **Busy** appears

- **Busy** appears in the **CNSLabSvrDC** icon
- Wait 5 minutes and then click on **Refresh** icon. WARNING: Do not click too early.
- The button should now be grey

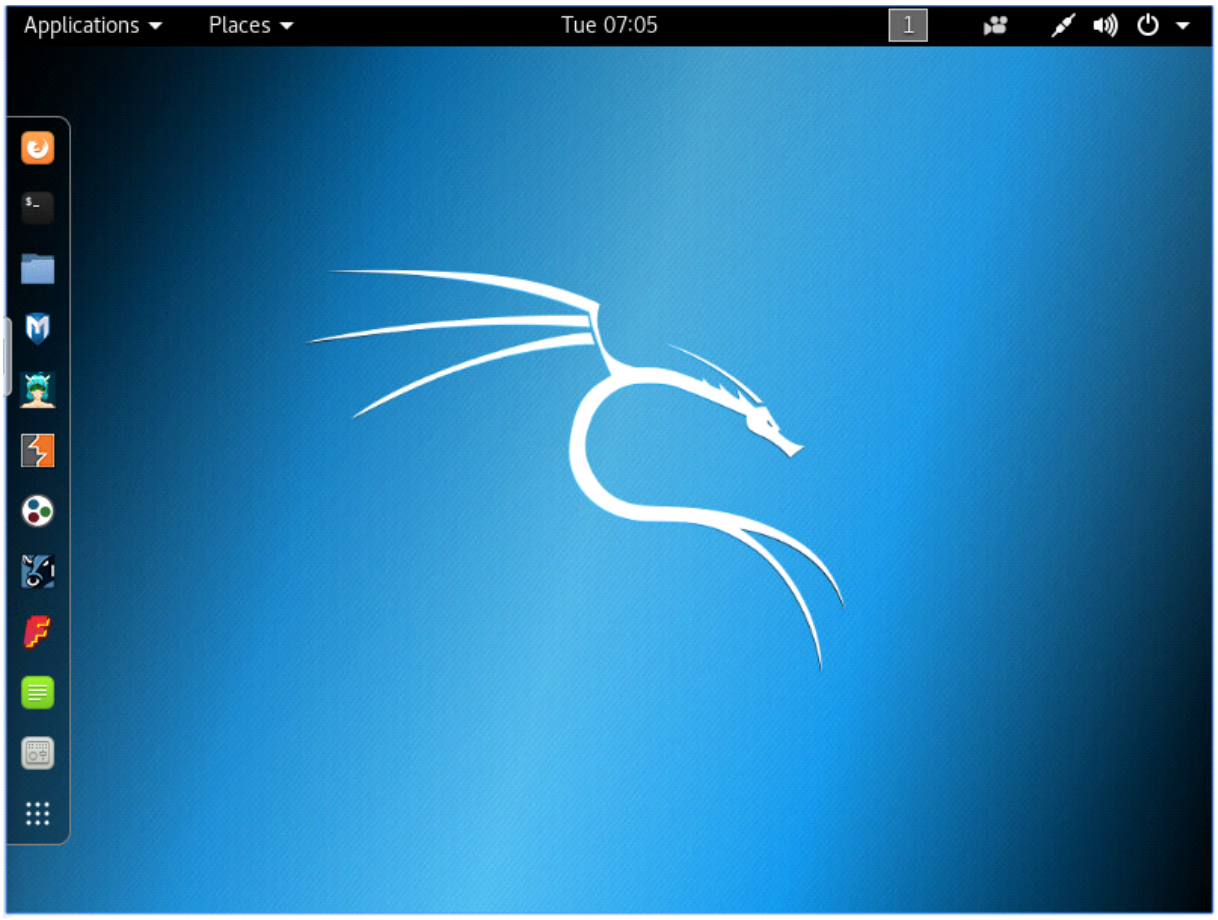


Session Timeout

- If the session is idle too long, the following screen will appear
- To return to the session, click anywhere in the window
- Press **Enter**



- The login screen appears
- Enter **toor** (root backwards) for the password. Press **Enter**
- The graphical desktiop appears where you left off



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